Class 10 NCERT Maths

Chapter 1: Real Numbers (Core-Basics Edition)

A simplified, detailed guide in Hinglish
Special edition for students who skipped Class 7-8-9
Core basics (RED) ko miss mat karna!

IS PDF MEIN RED RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).
Ye woh cheezein hain jo tujhe pehle se aani chahiye thi.
Inhe DHYAAN se padh - inke bina chapter samajh nahi aayega.

Agle page pe SAARE core basics ek saath diye hain.
Phir asli Chapter 1 shuru hota hai.

CORE BASICS - MISS MAT KARNA (RED PAGE)

Ye page sirf un cheezon ka hai jo Class 7-8-9 mein
padhayi jaati hain. 10th ka Chapter 1 inhi pe khada hai.

1) FACTOR KYA HOTA HAI?
Factor = jo number kisi doosre number ko POORA
(bina remainder) divide kar de.
Example: 12 ke factors = 1, 2, 3, 4, 6, 12
(kyunki ye sab 12 ko poora divide karte hain)

2) MULTIPLE KYA HOTA HAI?
Multiple = number ki "table" ke numbers.
Example: 3 ke multiples = 3, 6, 9, 12, 15, ...
(Factor chhota hota hai, multiple bada.)

3) PRIME, COMPOSITE, AUR 1

PRIME = sirf 2 factors (1 aur khud). Jaise 2,3,5,7,11
COMPOSITE = 2 se zyada factors. Jaise 4,6,8,9,10
1 NA prime hai NA composite (sirf 1 factor hai).
2 ekloti EVEN prime number hai.

4) POWER / EXPONENT (chhota upar wala number)
2^3 ka matlab 2 ko 3 baar multiply = 2x2x2 = 8.
Yahan 2 = base, 3 = power/exponent.
a^2 ko "a square", a^3 ko "a cube" bolte hain.

5) SQUARE ROOT (sqrt ya the root symbol)
Square root = power ka ULTA.
sqrt(9) = 3 kyunki 3x3 = 9.
sqrt(25) = 5 kyunki 5x5 = 25.
Symbol: tick-jaisa nishaan (root sign) number ke upar.
Is PDF mein hum "sqrt(n)" likhte hain (jaise sqrt2).

6) p/q MEIN q ZERO KYUN NAHI HO SAKTA?
p/q ka matlab p ko q se DIVIDE karna.
Kisi cheez ko 0 se divide karna ALLOWED nahi
(5/0 ka koi answer exist nahi karta - undefined).
Isliye rational number p/q mein hamesha q != 0.

7) HCF (Highest Common Factor)
HCF = do numbers ka sabse BADA common factor.
12 ke factors: 1,2,3,4,6,12
18 ke factors: 1,2,3,6,9,18
Common: 1,2,3,6 -> sabse bada = 6 -> HCF = 6

8) LCM (Least Common Multiple)
LCM = do numbers ka sabse CHHOTA common multiple.
12 ke multiples: 12,24,36,48...
18 ke multiples: 18,36,54...
Common sabse chhota = 36 -> LCM = 36

9) NUMBER TYPES (jaldi recap)
Natural (N): 1,2,3,... Whole (W): 0,1,2,3,...
Integers (Z): ...-2,-1,0,1,2...
Rational (Q): p/q form (q!=0), jaise 1/2, 0.25
Irrational: p/q mein nahi aata, jaise sqrt2, pi
Real (R): rational + irrational dono.

10) TERMINATING vs NON-TERMINATING DECIMAL
Terminating = decimal khatam ho jaye: 0.5, 0.25
Non-terminating repeating = pattern repeat: 0.333...
Non-terminating non-repeating = na khatam na pattern:
sqrt2 = 1.41421356... (ye IRRATIONAL hota hai)

NOW: ASLI CHAPTER 1 SHURU

TOPIC 1: FUNDAMENTAL THEOREM OF ARITHMETIC

STATEMENT:

"Har composite number ko prime numbers ke product
(multiplication) ke roop mein likha ja sakta hai,
aur ye tareeka UNIQUE hota hai (order chhod ke)."

[CORE] "Product of primes" matlab prime numbers ko multiply karna. Pehle prime/factor wala page dekh.

EXAMPLE 1: 156 ka prime factorisation

156 / 2 = 78
78 / 2 = 39
39 / 3 = 13
13 / 13 = 1

156 = 2 x 2 x 3 x 13 = 2^2 x 3 x 13

[CORE] 2^2 ka matlab 2x2 (power wala page yaad kar).

EXAMPLE 2: 3825 ka prime factorisation

3825 / 3 = 1275
1275 / 3 = 425
425 / 5 = 85
85 / 5 = 17
17 / 17 = 1

3825 = 3^2 x 5^2 x 17

TOPIC 2: HCF AND LCM BY PRIME FACTORISATION

METHOD:

HCF = common prime factors ka product (LOWEST power)
LCM = saare prime factors ka product (HIGHEST power)

[CORE] HCF/LCM ka basic matlab RED page pe diya hai (point 7 aur 8). Pehle wo samajh, phir ye method.

EXAMPLE 3: 96 aur 404 ka HCF aur LCM

96 = 2^5 x 3
404 = 2^2 x 101

HCF = 2^2 = 4 (common prime 2, lowest power)
LCM = 2^5 x 3 x 101 = 9696 (all primes, highest power)

Verify: HCF x LCM = 4 x 9696 = 38784
96 x 404 = 38784 (match!)

[CORE] Sirf 2 numbers ke liye: HCF x LCM = product of the two numbers. (3 numbers pe ye rule nahi chalta.)

TOPIC 3: IRRATIONAL NUMBERS (sqrt2 is irrational)

[CORE] sqrt(2) ka matlab "woh number jisko square karne pe 2 aaye". Root symbol RED page point 5 mein.

THEOREM USED:

Agar prime p, a^2 ko divide karta hai, toh p, a ko bhi divide karta hai.

PROOF: sqrt2 is irrational (Proof by Contradiction)

Step 1: Maan le sqrt2 rational hai. Toh sqrt2 = p/q
where q != 0 and HCF(p,q) = 1 (simplest form).

[CORE] q != 0 kyun? RED page point 6 dekh.
HCF(p,q)=1 matlab fraction simplest form mein.

Step 2: Square dono side:
2 = p^2 / q^2 -> 2 q^2 = p^2 ...(i)

Step 3: 2 divides p^2 -> 2 divides p.
Step 4: So p = 2c.
Step 5: (i) mein daal: 2 q^2 = 4 c^2 -> q^2 = 2 c^2
Step 6: 2 divides q^2 -> 2 divides q.
Step 7: Ab p aur q dono ko 2 divide karta hai, par
humne kaha tha HCF = 1. CONTRADICTION!

CONCLUSION: Assumption galat. sqrt2 IRRATIONAL hai.

QUICK RESULTS (yaad rakh):
- sqrt(p) irrational hota hai jab p prime ho.
- Rational + Irrational = Irrational
- (non-zero) Rational x Irrational = Irrational

Q AND A TIME

(Pehle khud try kar, phir mujhe bhej - main check karunga)

Q1. 140 ko prime factors ke product mein likho.

Q2. 26 aur 91 ka HCF aur LCM nikaal (prime factorisation se).

Q3. sqrt(5) ko irrational prove kar (sqrt2 wala method copy kar).

Q4. CORE CHECK: 1 prime hai ya composite? Kyun?

Q5. CORE CHECK: 5/0 ka answer kya hai? p/q mein q ke baare mein kya rule hai?

Q6. 3 + 2 x sqrt(5) ko irrational dikhao.

SUMMARY

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1. Fundamental Theorem: har composite number = unique product of primes.

2. HCF = common primes x lowest power.
   LCM = all primes x highest power.

3. 2 numbers ke liye: HCF x LCM = product of numbers.

4. sqrt(p) irrational hai jab p prime ho.

5. Rational + Irrational = Irrational.

CORE BASICS (RED page) ek baar aur revise kar lena -
factor, multiple, prime, power, root, q!=0, HCF, LCM.
Inke bina ye chapter adhoora rahega.

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